

Engineering Curves: Construction of Cycloid

Rolling Circle Method with Tangent & Normal

Gajavada Sanjeevkumar

Problem Statement

Question

Draw a **cycloid** generated by a circle of diameter $D = 50$ mm rolling along a straight base line for **one complete revolution** without slipping.

Construct a **tangent** and a **normal** to the cycloid at any point P located on the curve.

Step 1: Draw Generating Circle & Directing Line

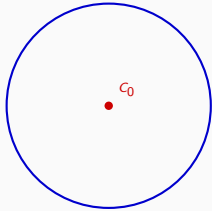
C_0 → Initial Center Point

AA' → Directing Line

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Step 1: Draw Generating Circle & Directing Line

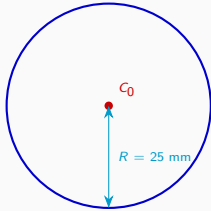


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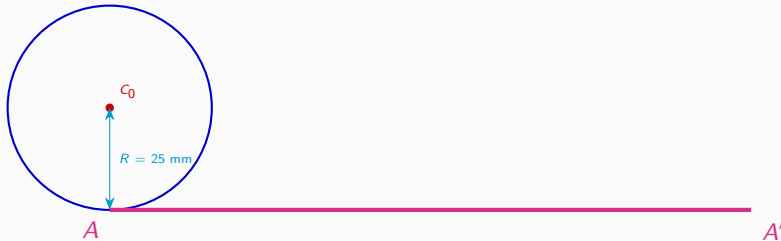
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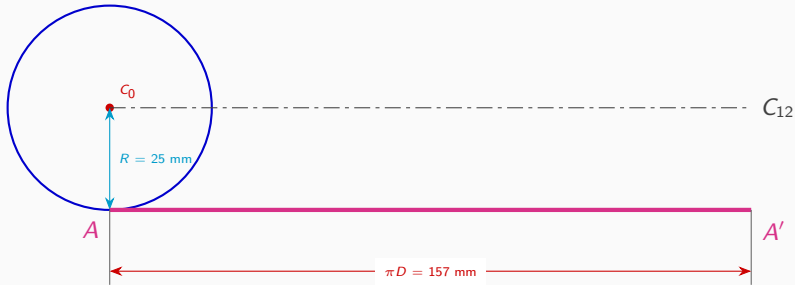
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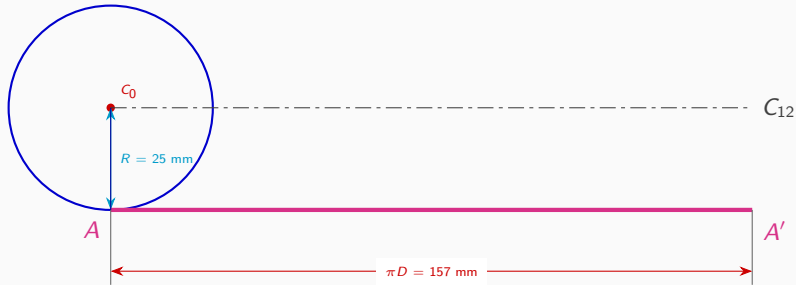
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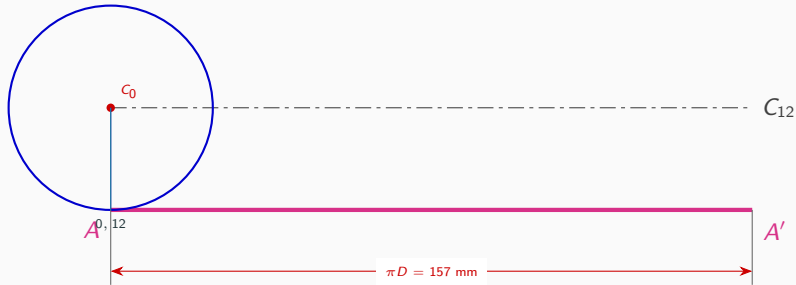
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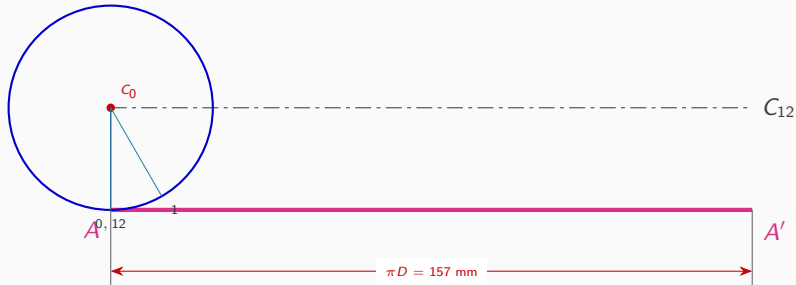
Step 2: Divide Generating Circle into 12 Equal Parts

AA' → Directing Line
Axis Line → Center Line
 $0 \dots 12$ → Circle Divisions



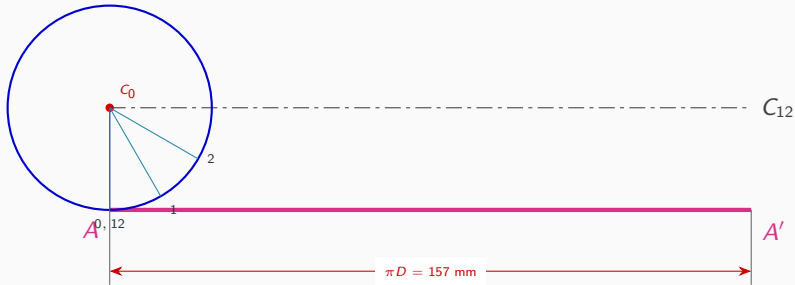
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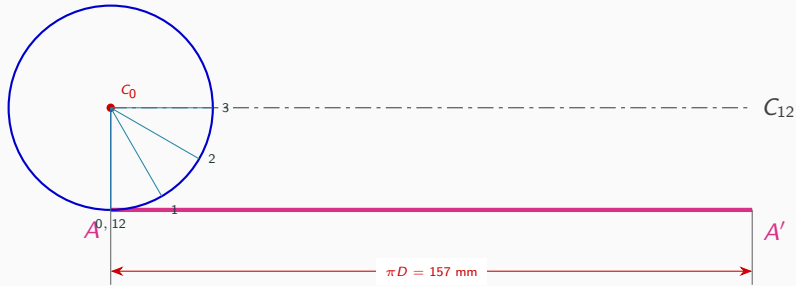
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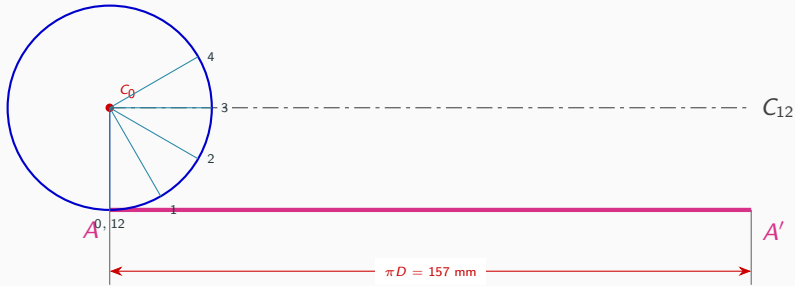
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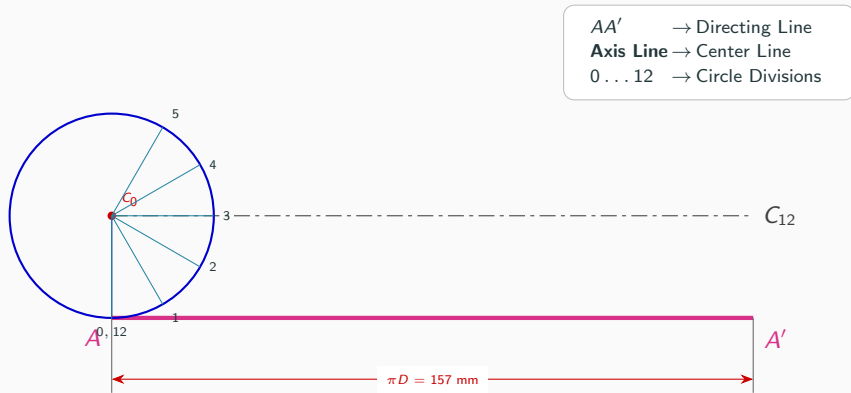


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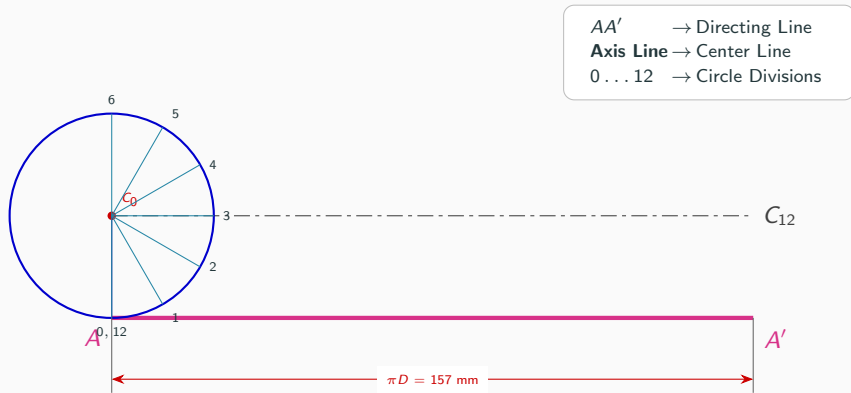
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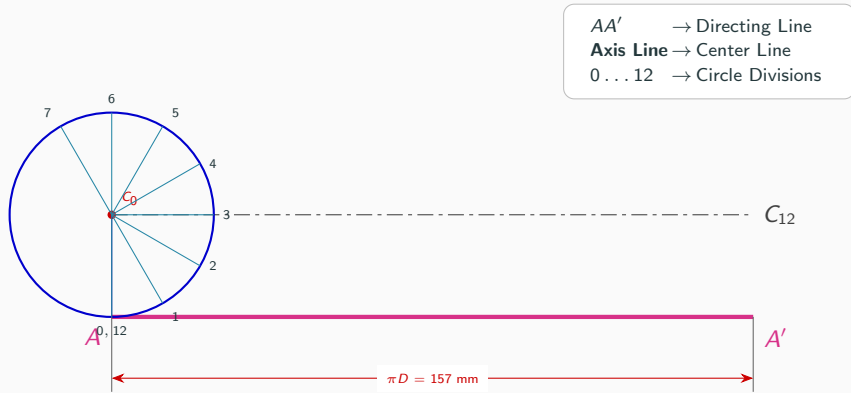
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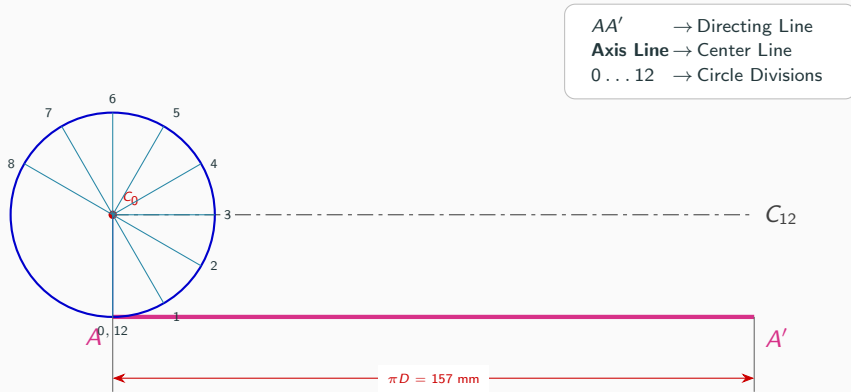
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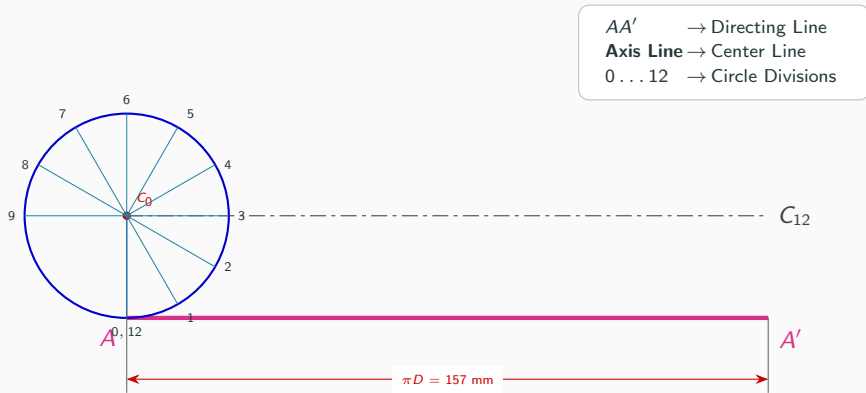
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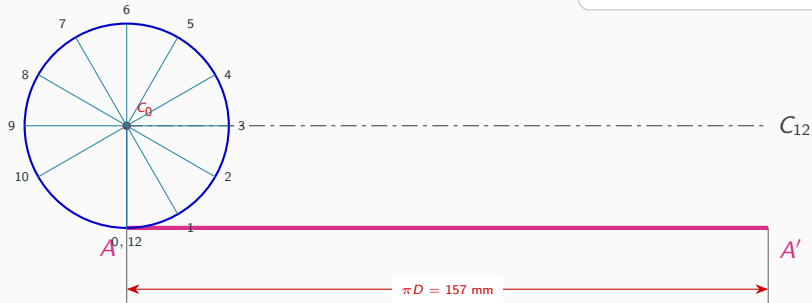
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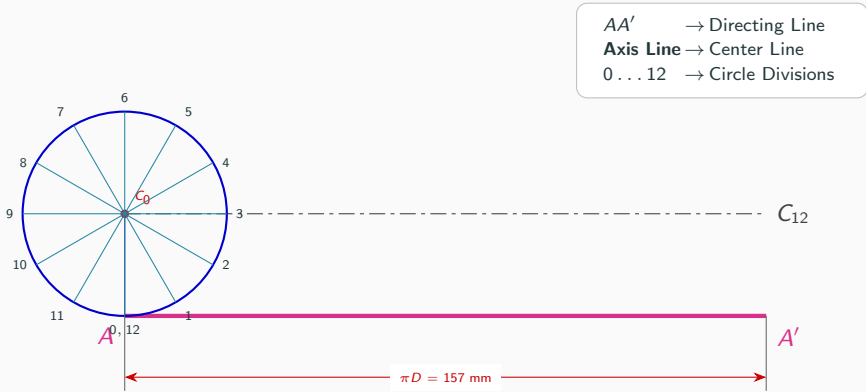
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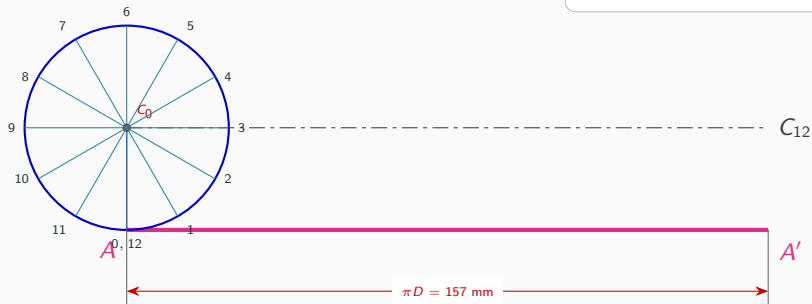
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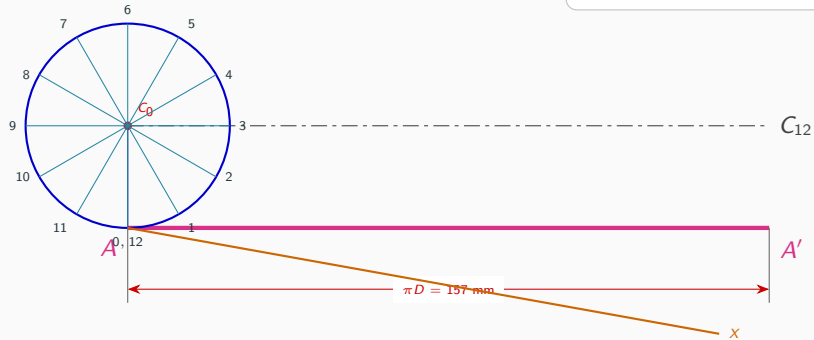
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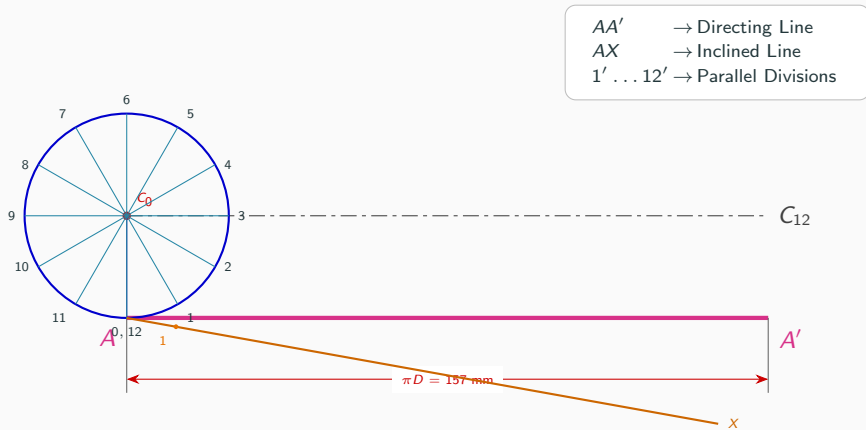
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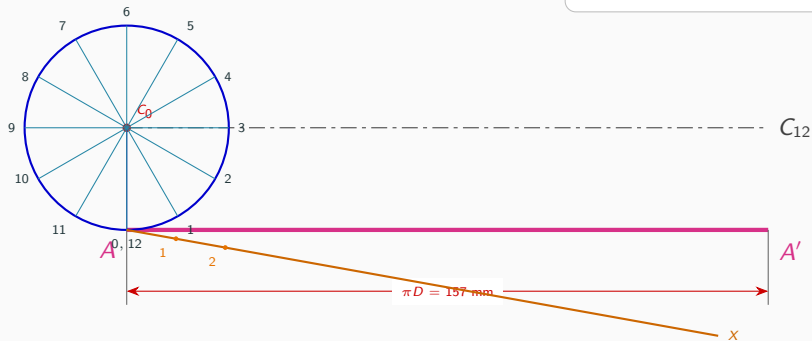


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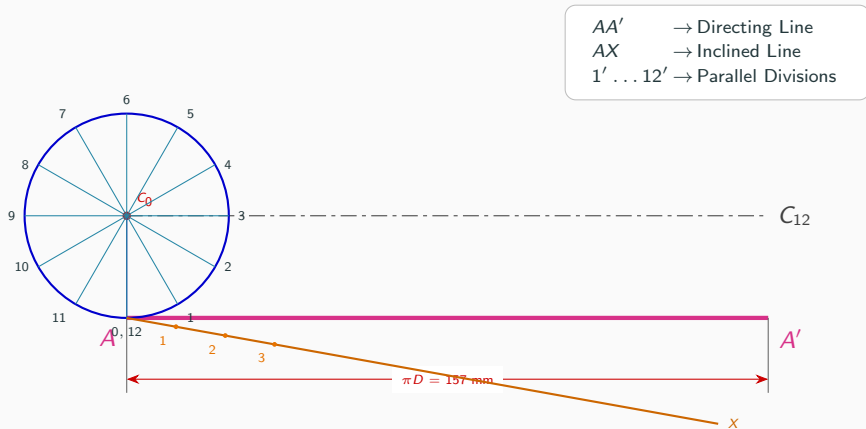


Step 3: Divide Directing Line using Line Division Method

AA' → Directing Line
 AX → Inclined Line
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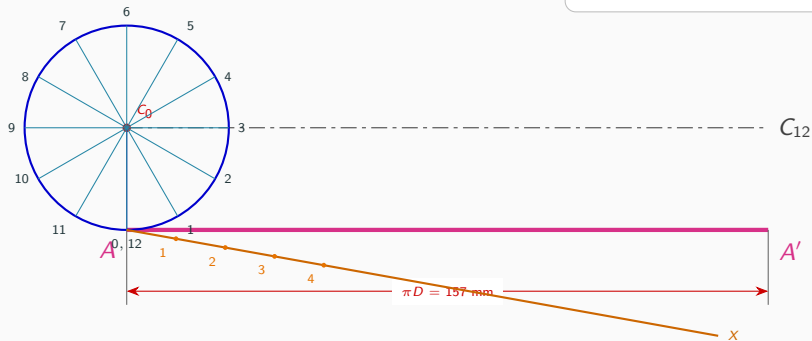


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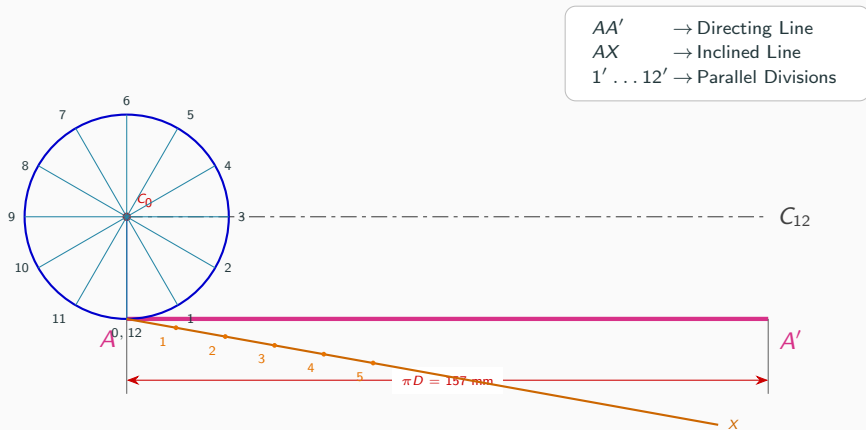


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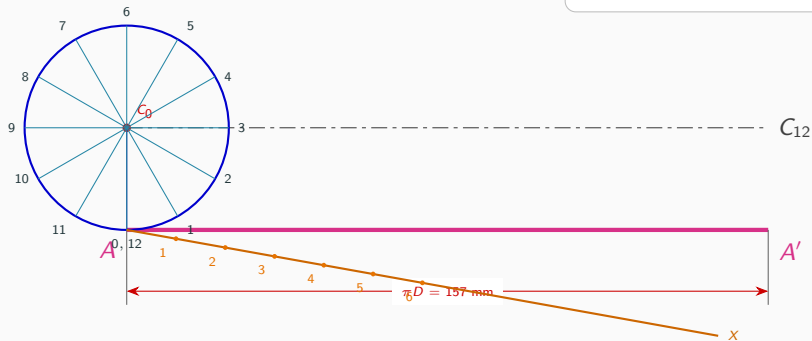


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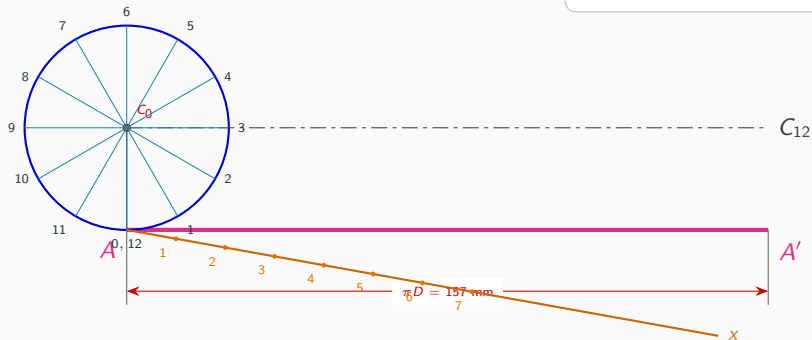
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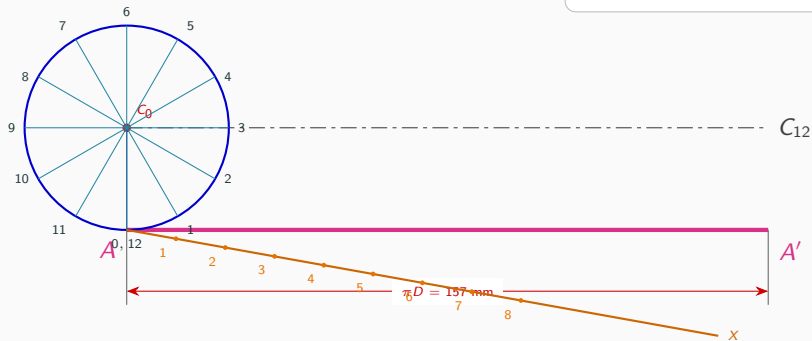
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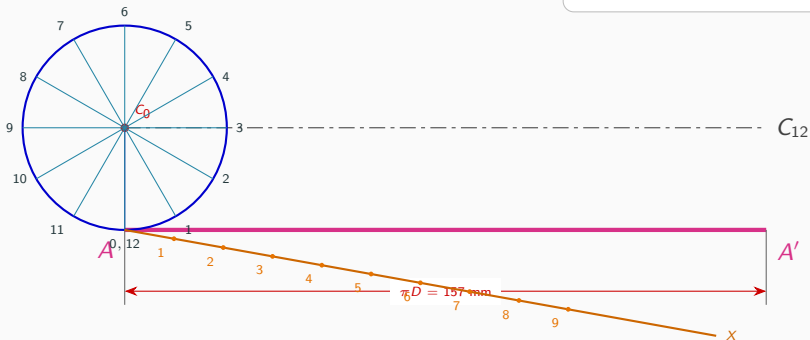
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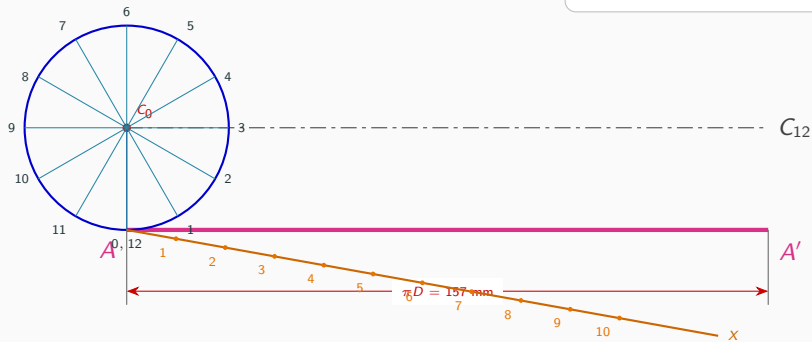
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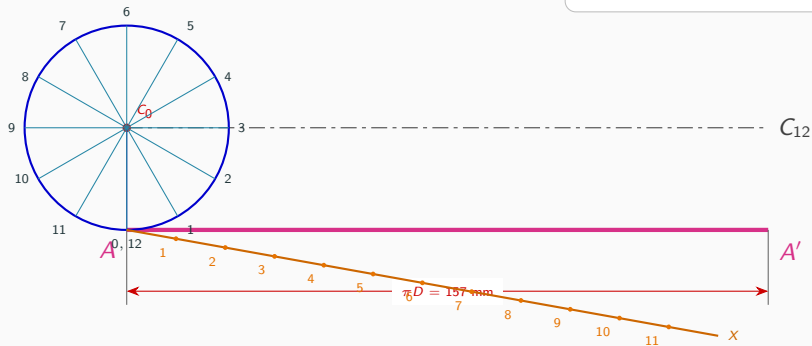
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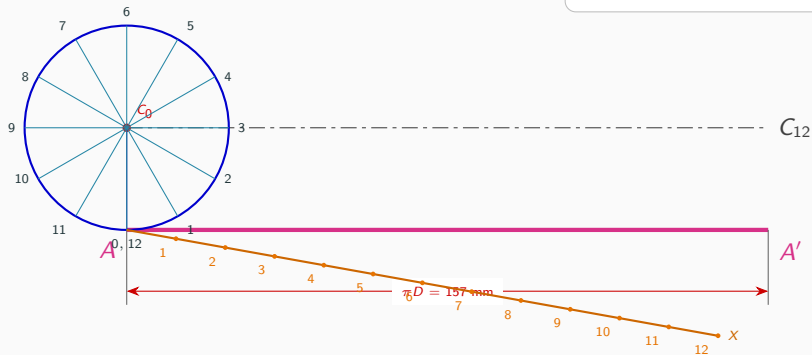
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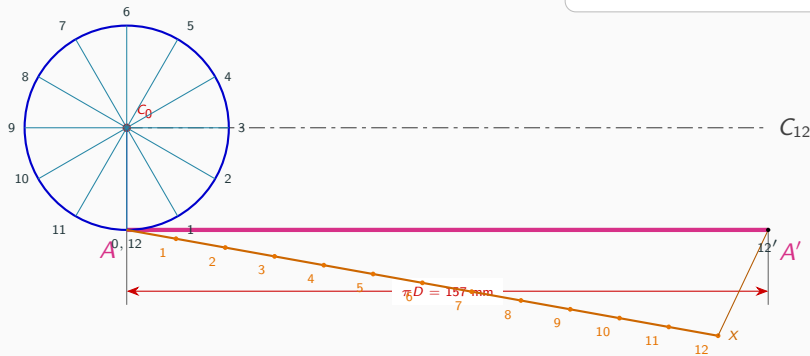
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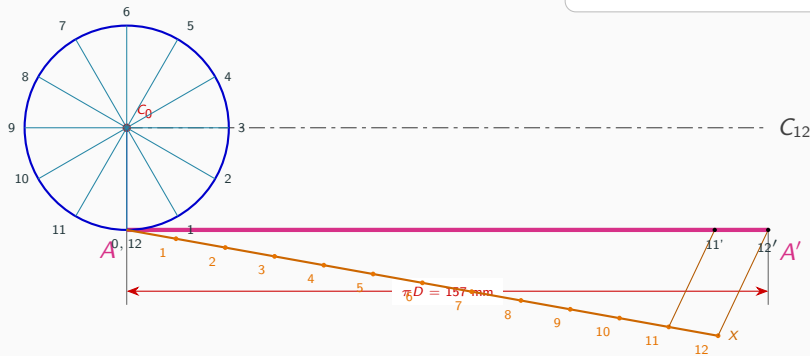
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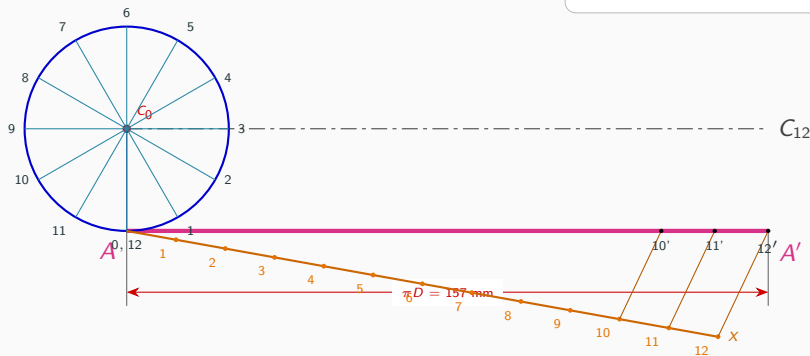
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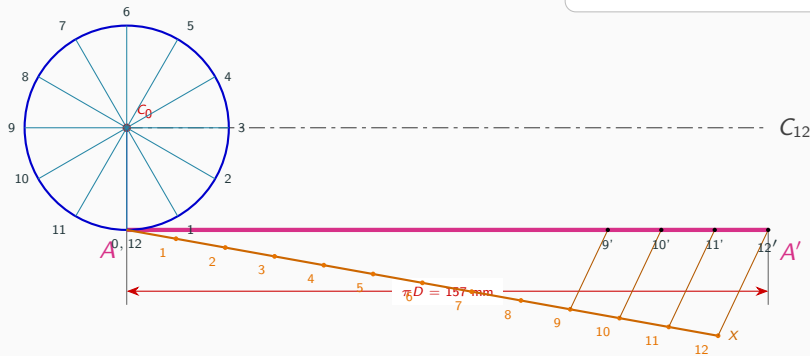
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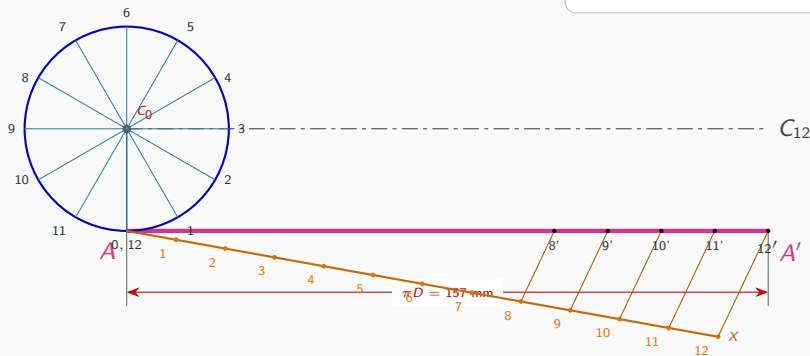
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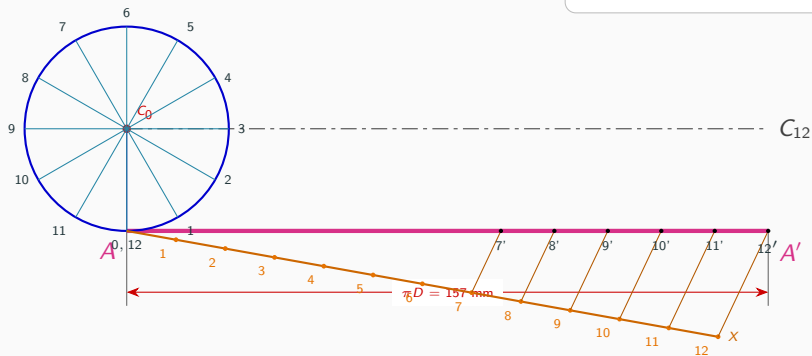
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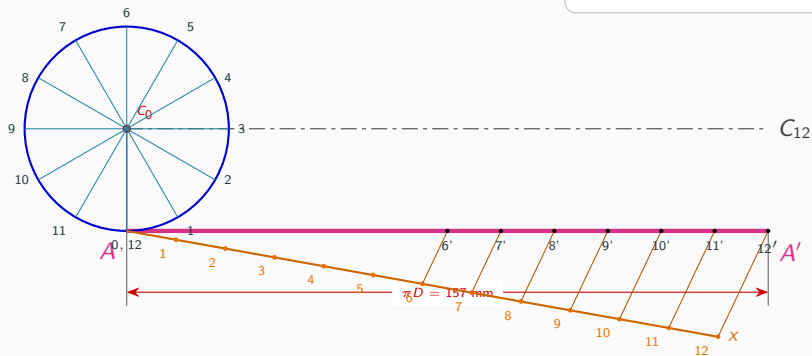
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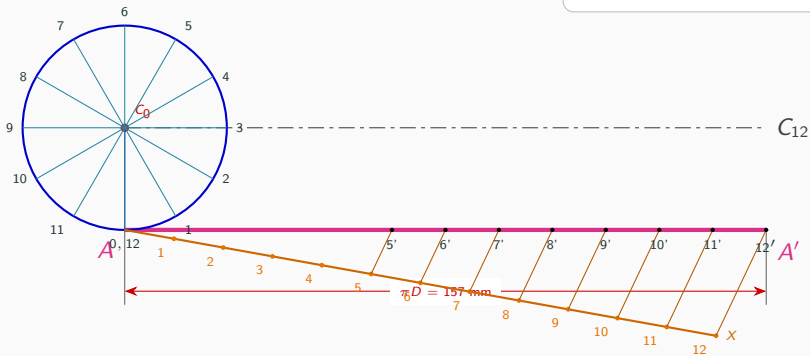
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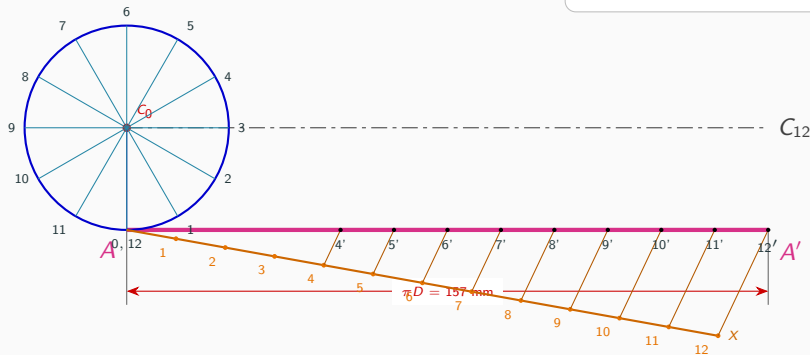
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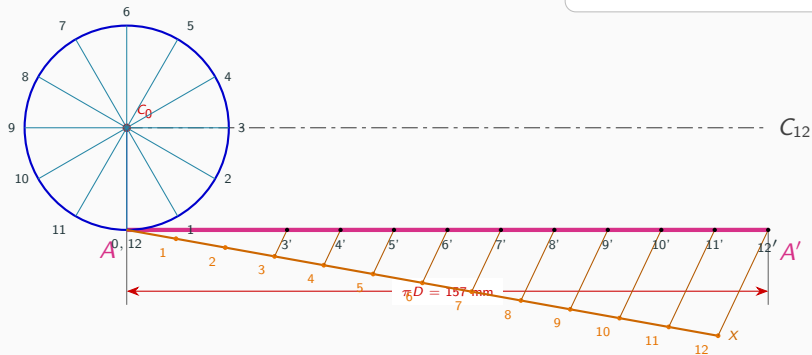
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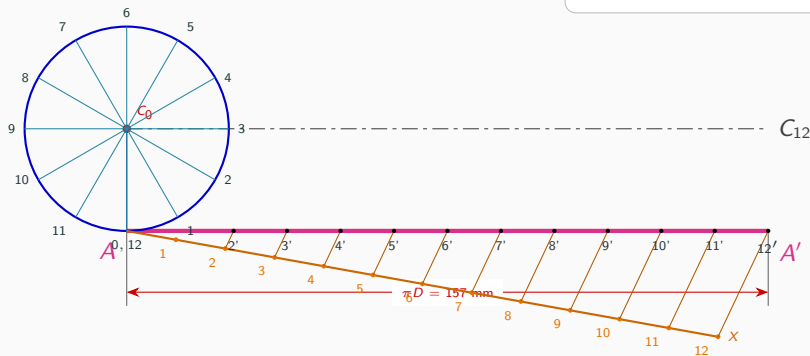
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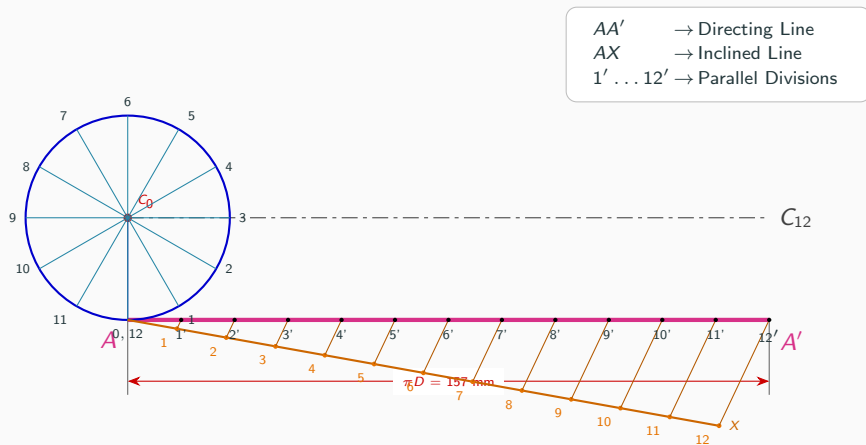


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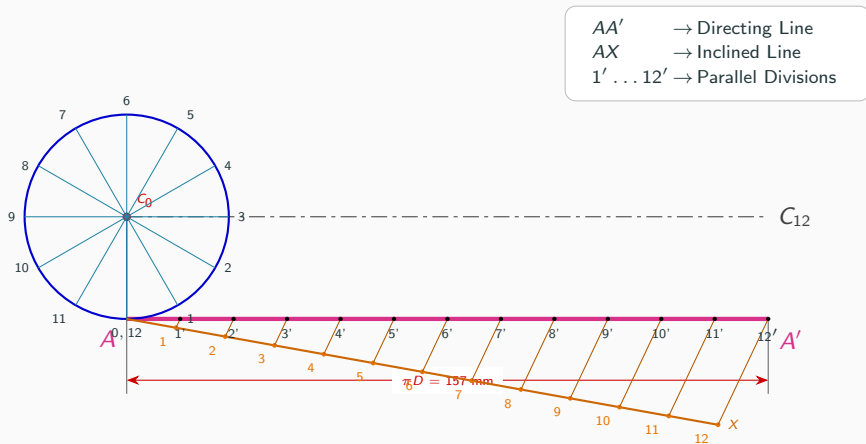
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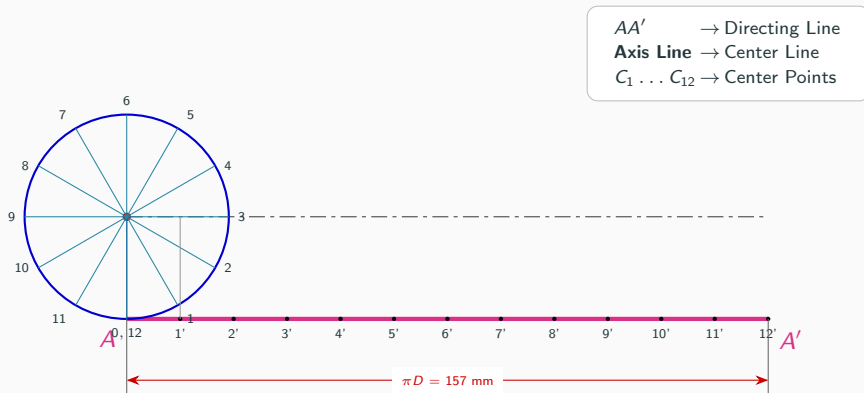
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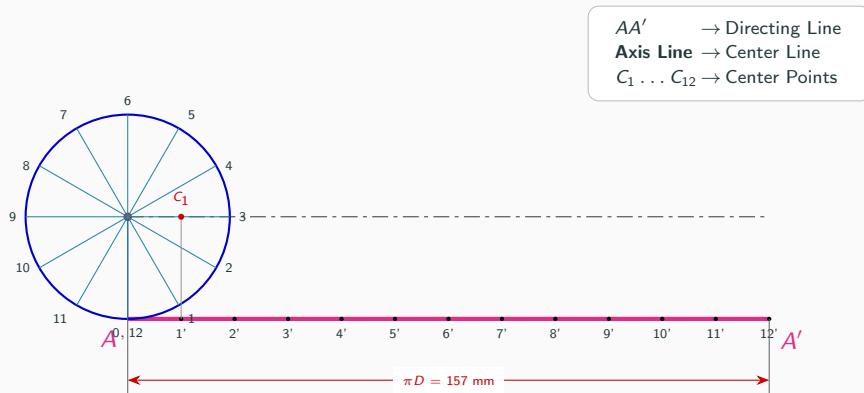
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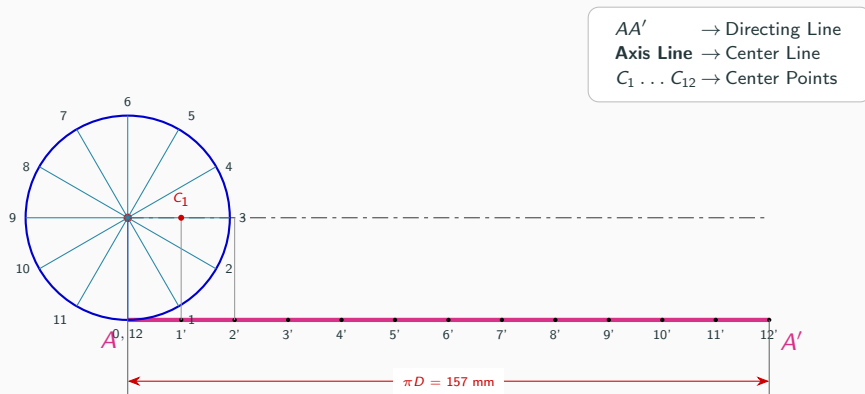
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



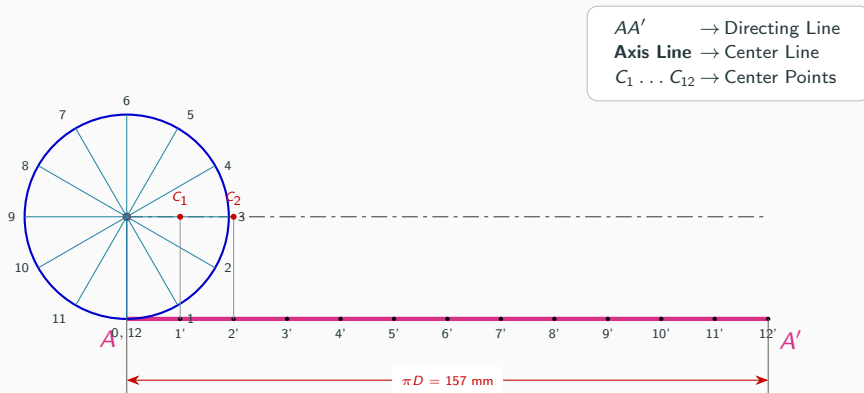
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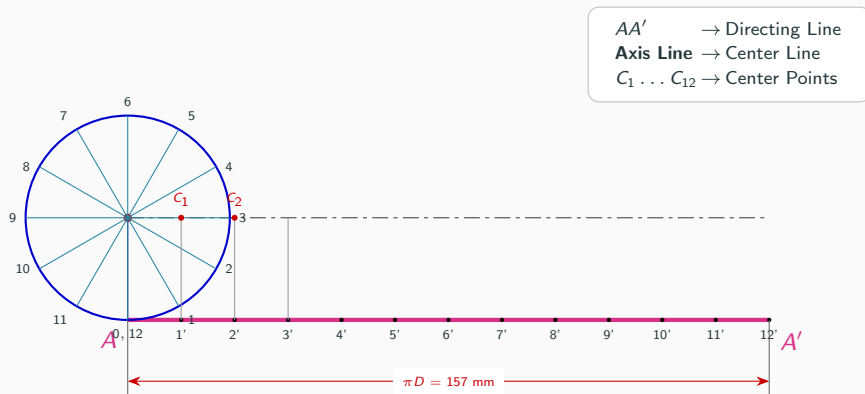
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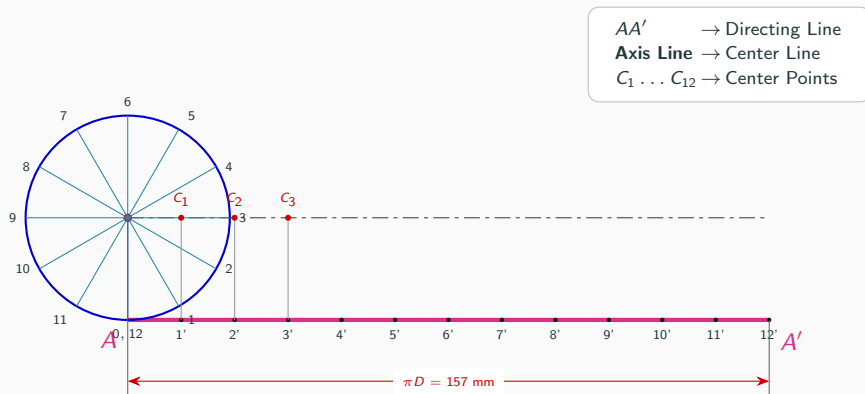
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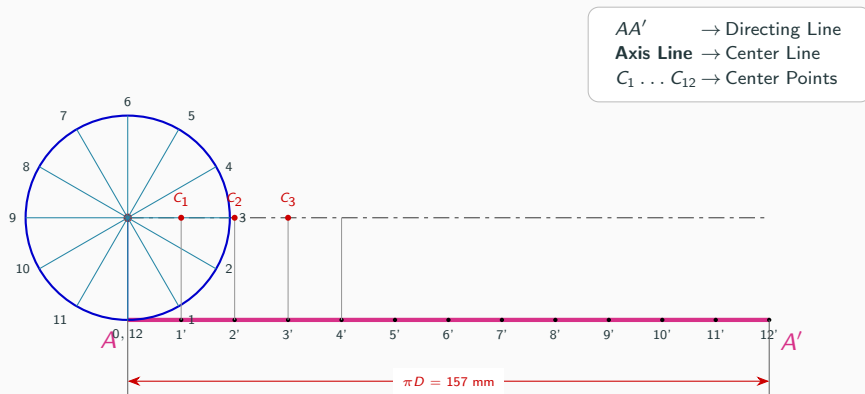
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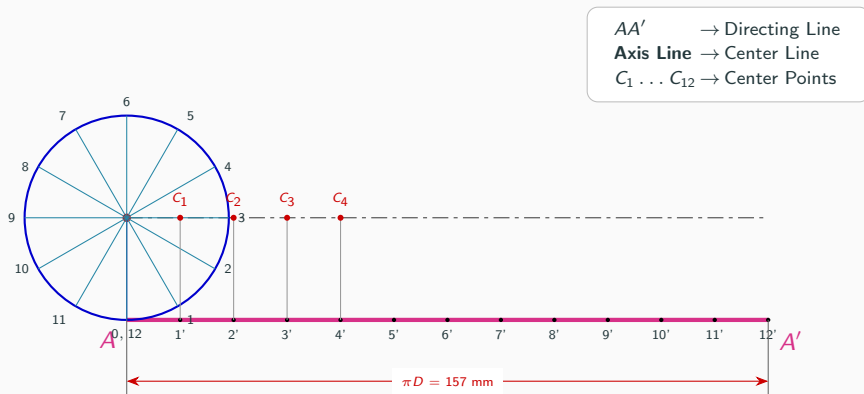
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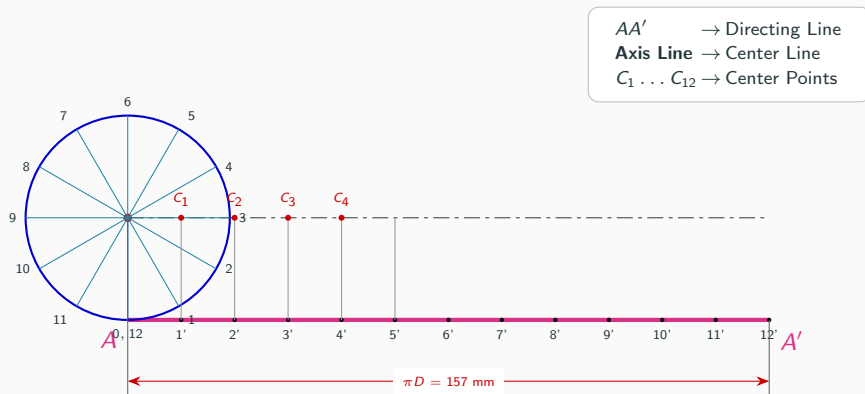
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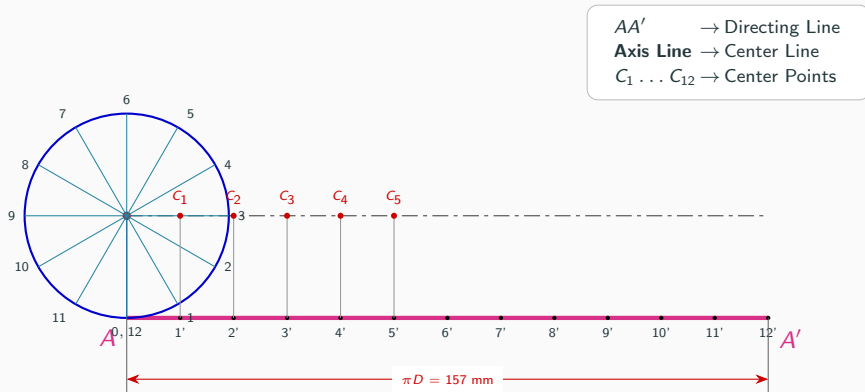
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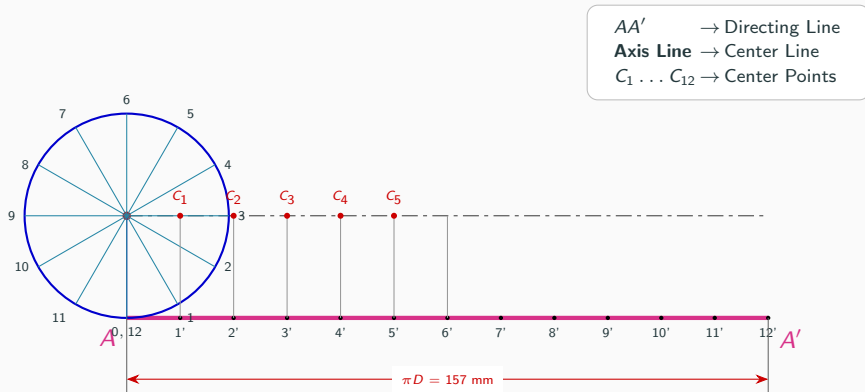
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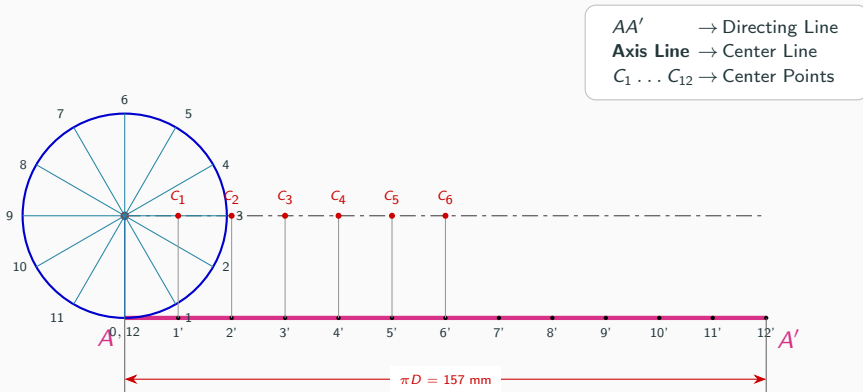
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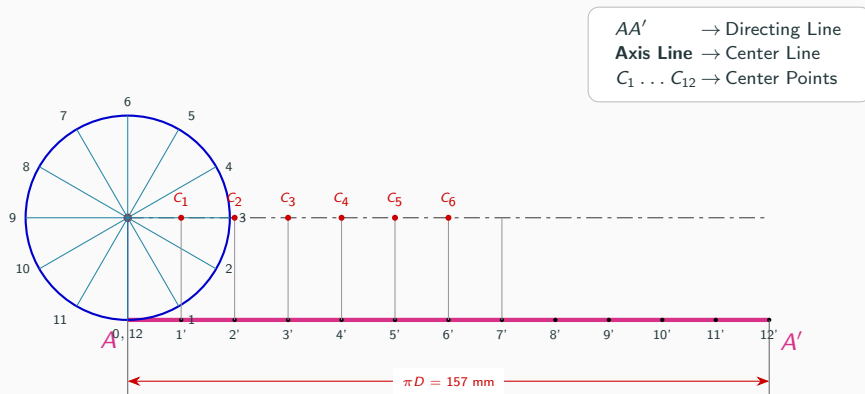
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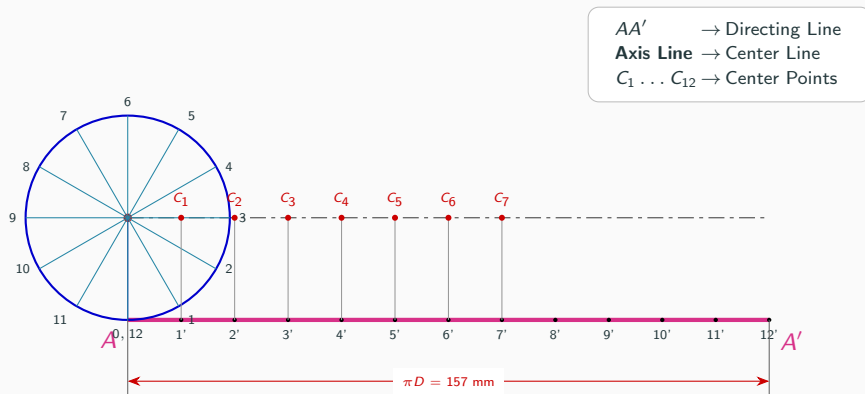
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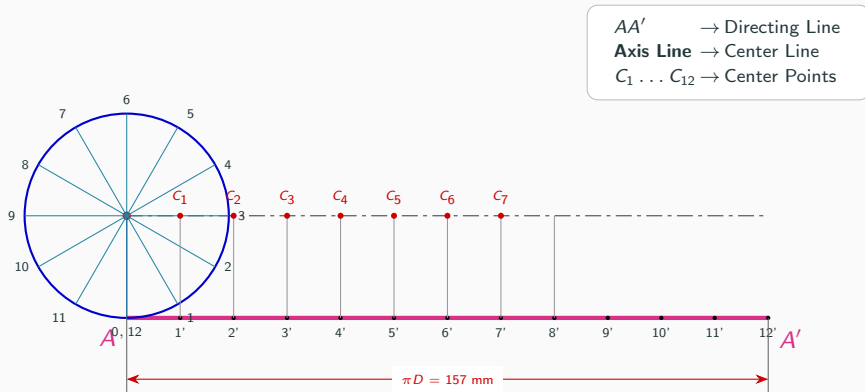
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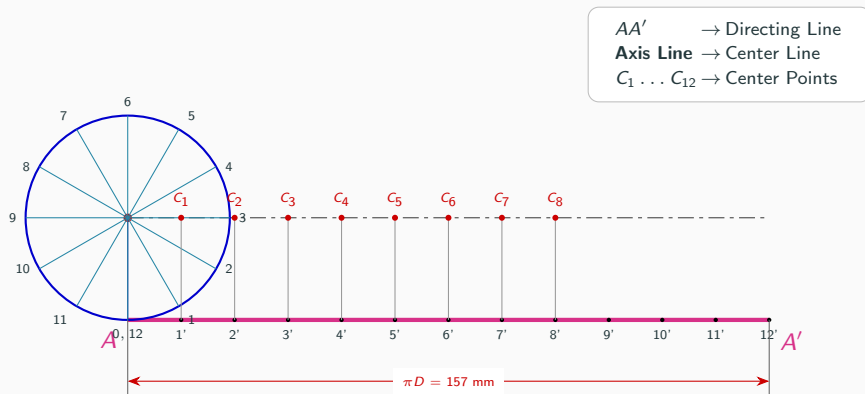
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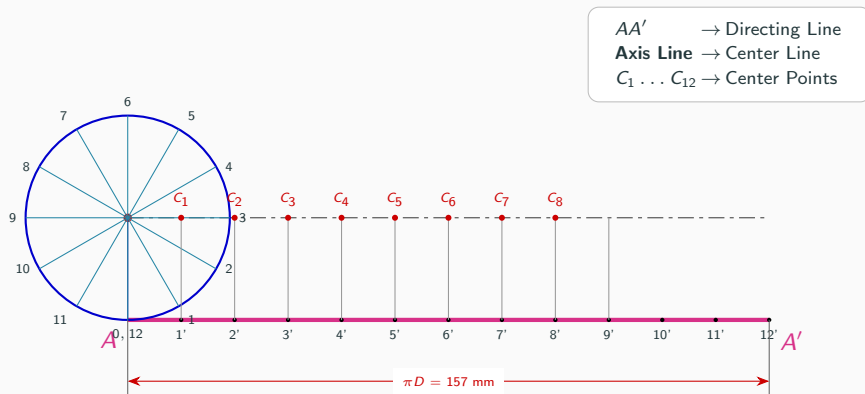
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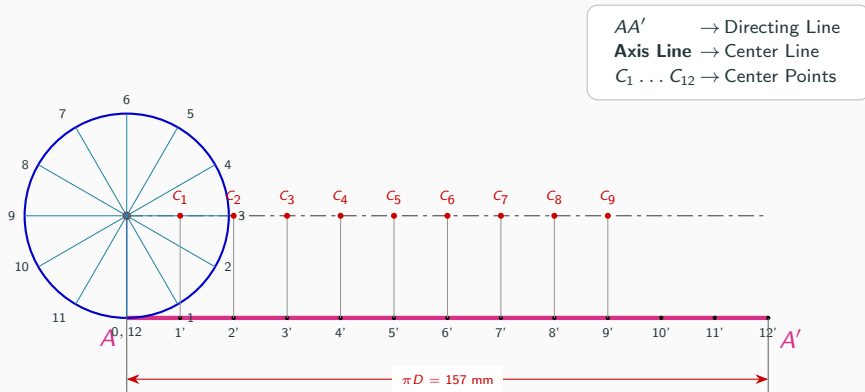
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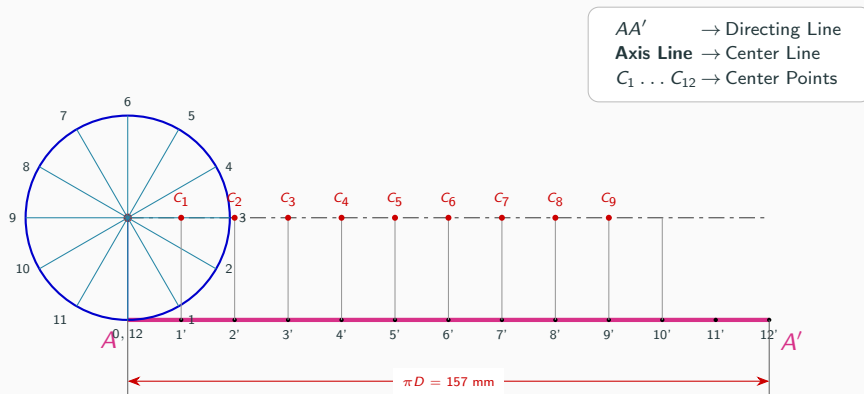
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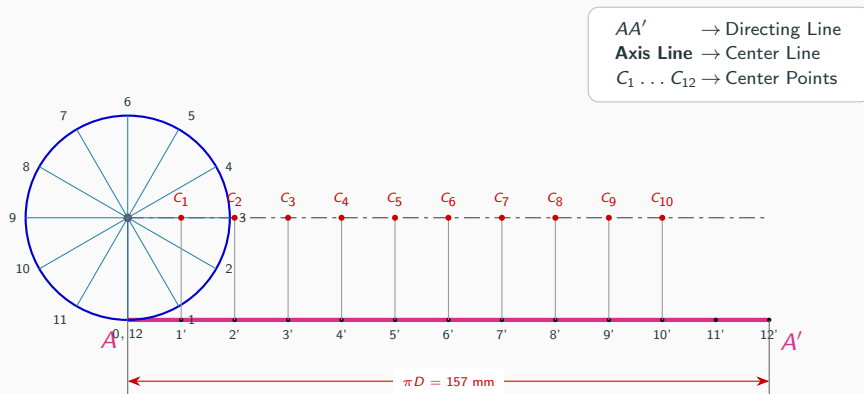
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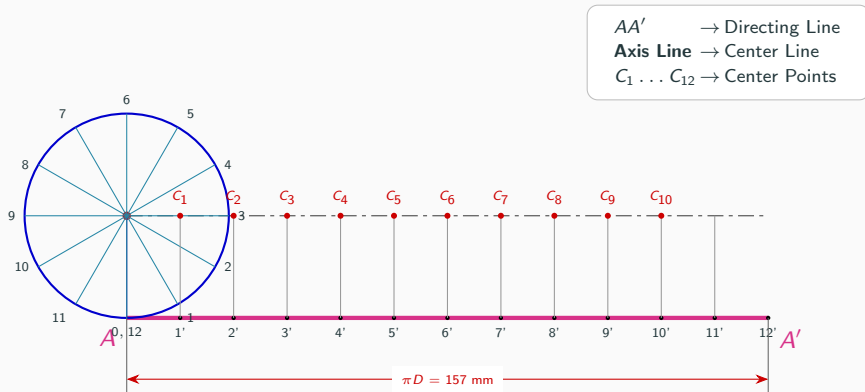
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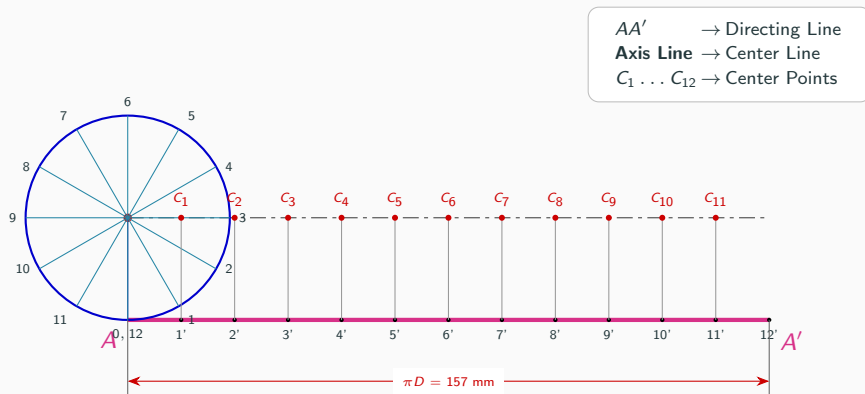
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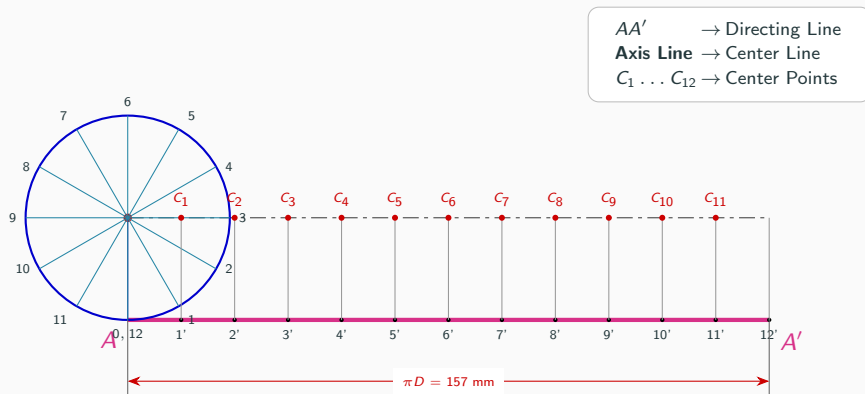
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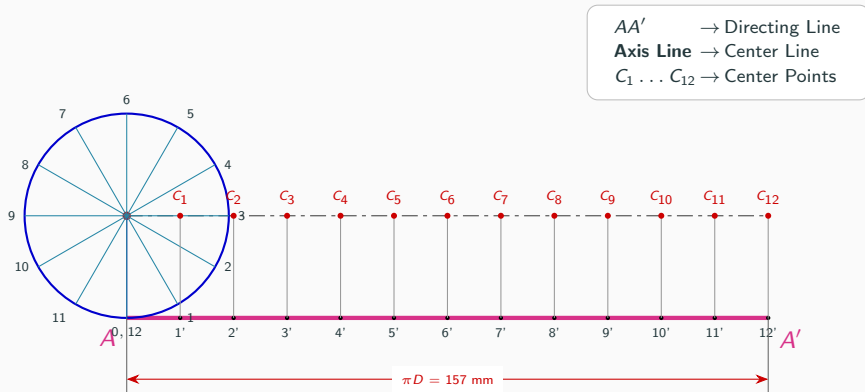
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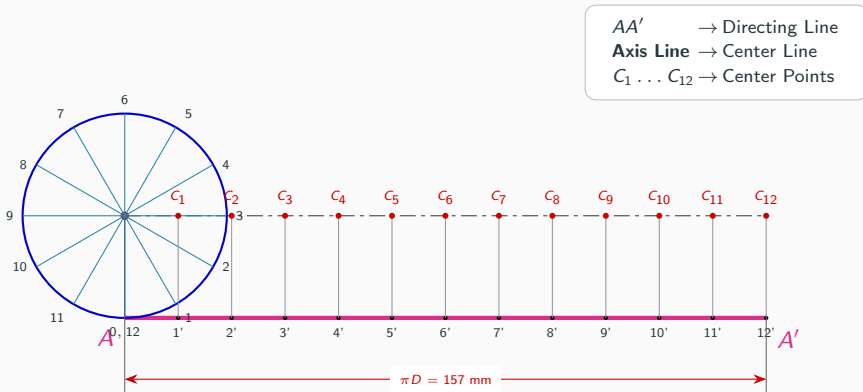
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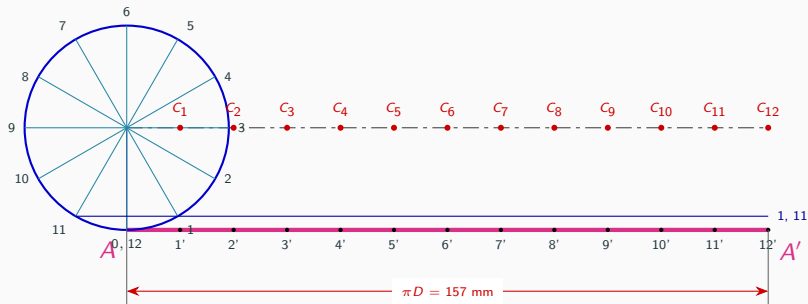
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Step 5: Draw Horizontal Reference Lines

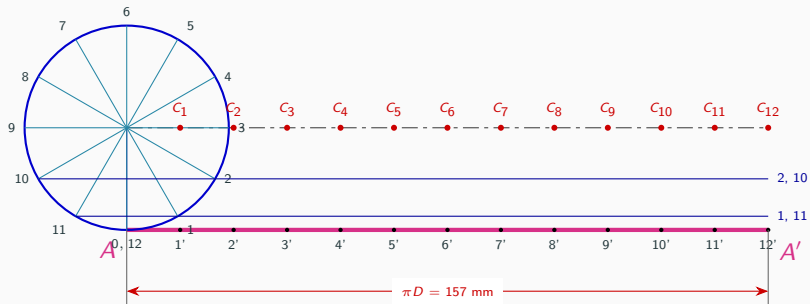
Blue Lines → Horizontal Reference Lines

$C_1 \dots C_{12}$ → Retained Centers



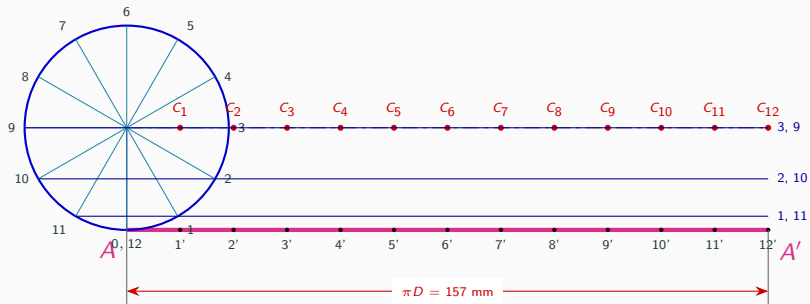
Step 5: Draw Horizontal Reference Lines

Blue Lines → Horizontal Reference Lines
 $C_1 \dots C_{12}$ → Retained Centers

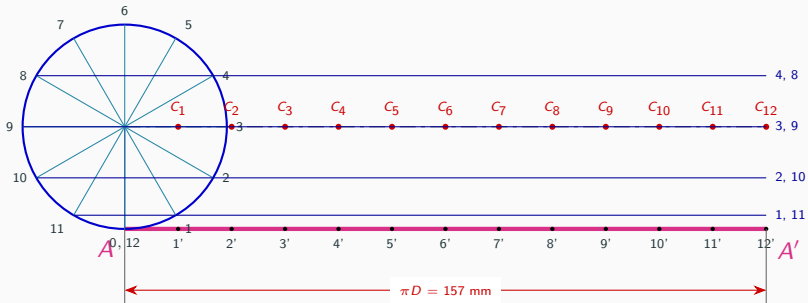


Step 5: Draw Horizontal Reference Lines

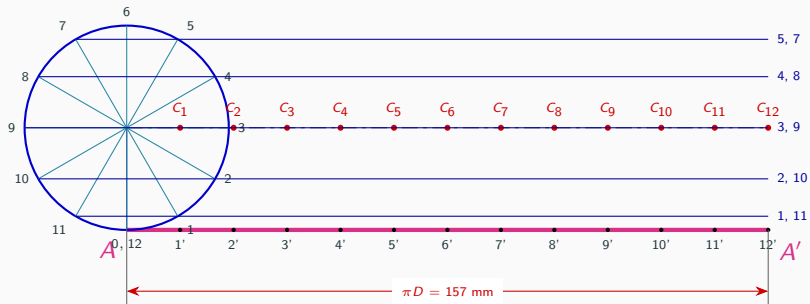
Blue Lines → Horizontal Reference Lines

$$C_1 \dots C_{12} \rightarrow \text{Retained Centers}$$


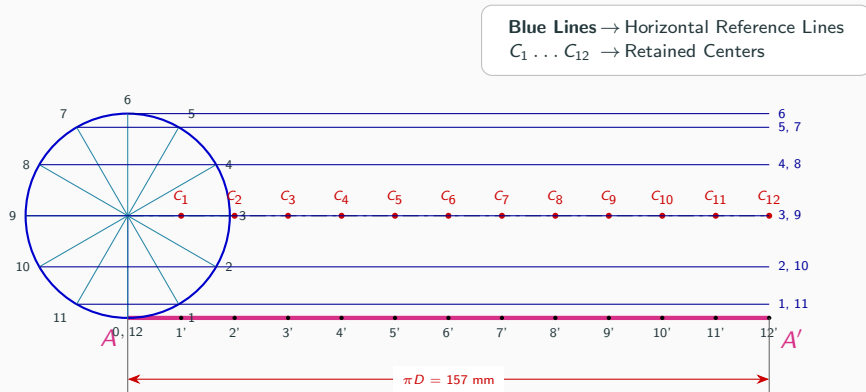
Step 5: Draw Horizontal Reference Lines

$$C_1 \dots C_{12} \rightarrow \text{Retained Centers}$$


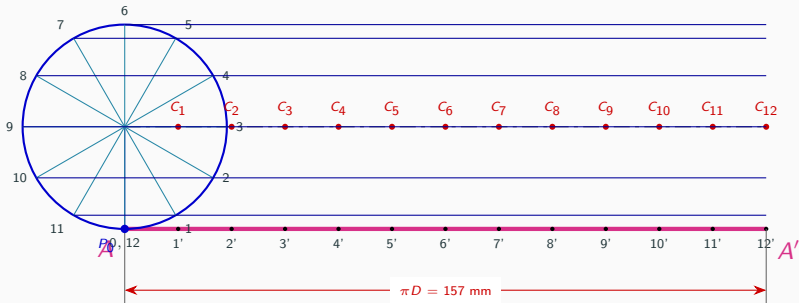
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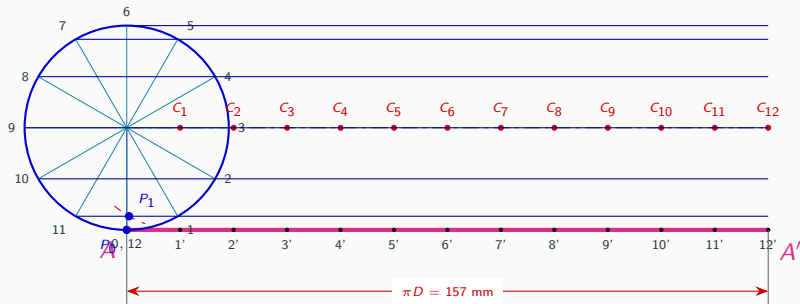


Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$

$$C_1 \dots C_{12} \rightarrow \text{Retained Centers}$$
$$P_1 \dots P_{12} \rightarrow \text{Cutting Arcs/Points}$$


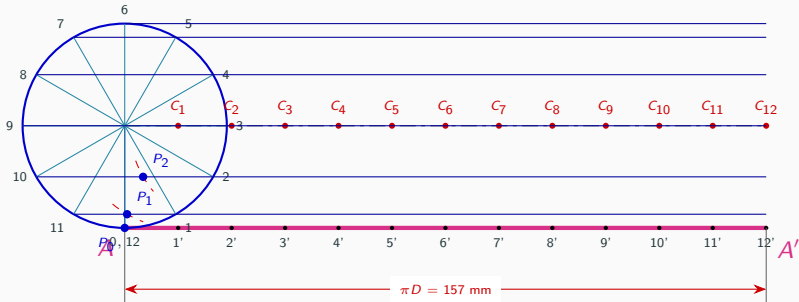
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$

$C_1 \dots C_{12} \rightarrow$ Retained Centers
 $P_1 \dots P_{12} \rightarrow$ Cutting Arcs/Points

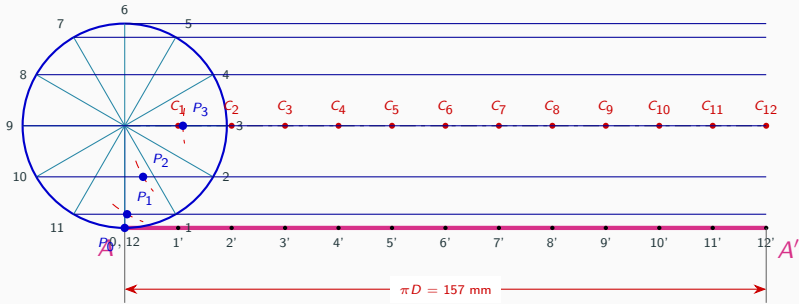


Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$

$C_1 \dots C_{12} \rightarrow$ Retained Centers
 $P_1 \dots P_{12} \rightarrow$ Cutting Arcs/Points

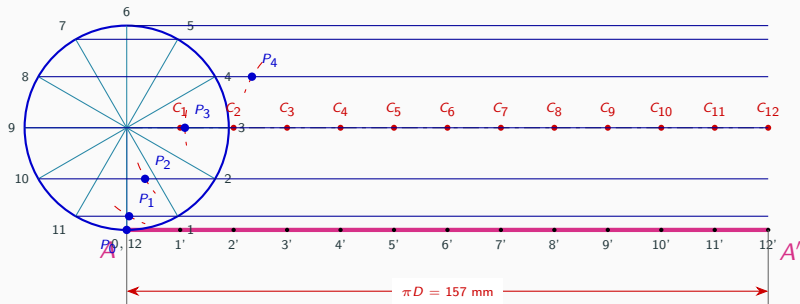


Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$

$$\begin{aligned} C_1 \dots C_{12} &\rightarrow \text{Retained Centers} \\ P_1 \dots P_{12} &\rightarrow \text{Cutting Arcs/Points} \end{aligned}$$


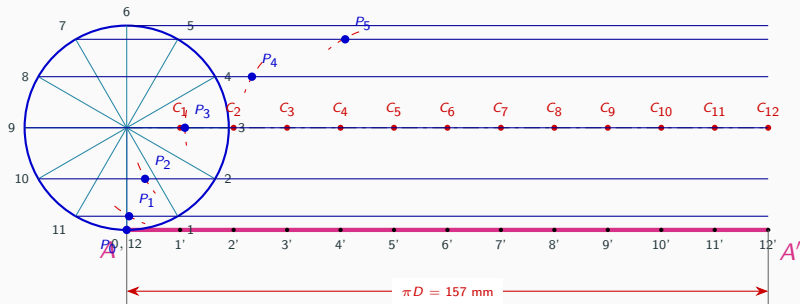
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$

$C_1 \dots C_{12} \rightarrow$ Retained Centers
 $P_1 \dots P_{12} \rightarrow$ Cutting Arcs/Points

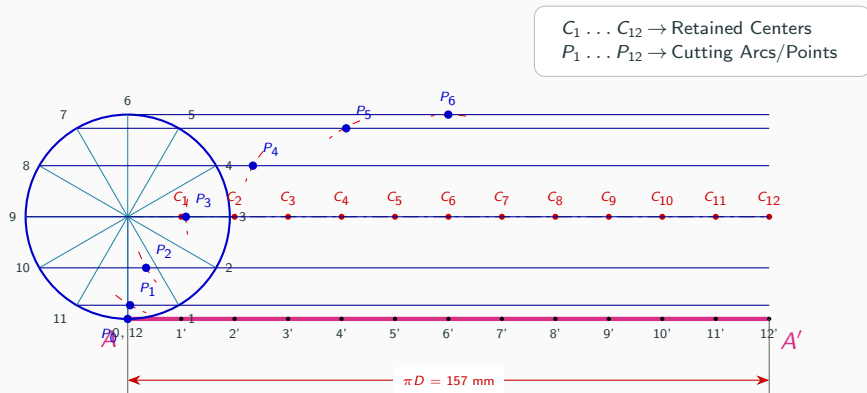


Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$

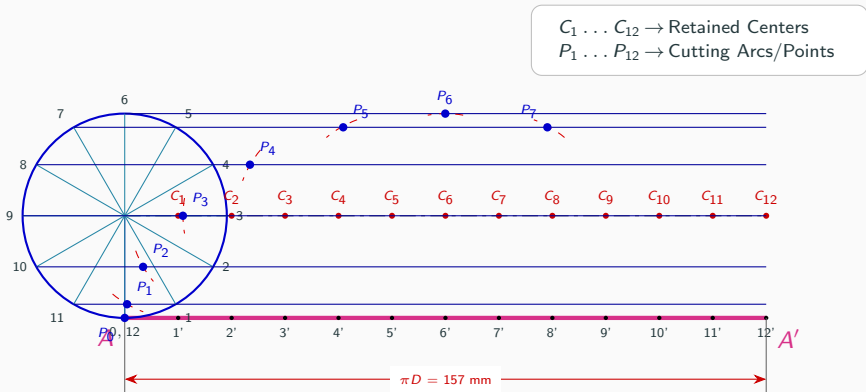
$C_1 \dots C_{12} \rightarrow$ Retained Centers
 $P_1 \dots P_{12} \rightarrow$ Cutting Arcs/Points



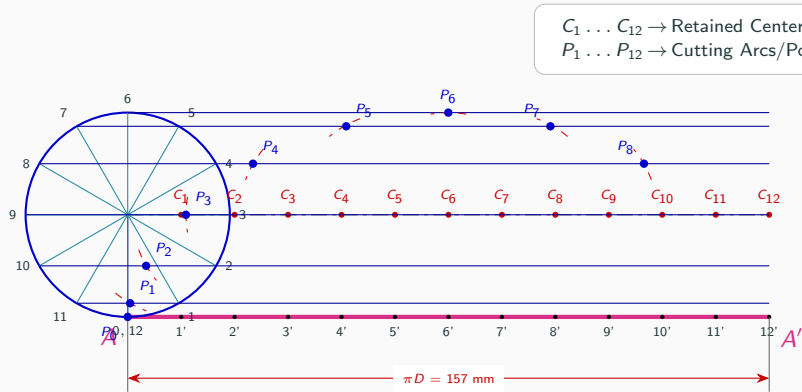
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



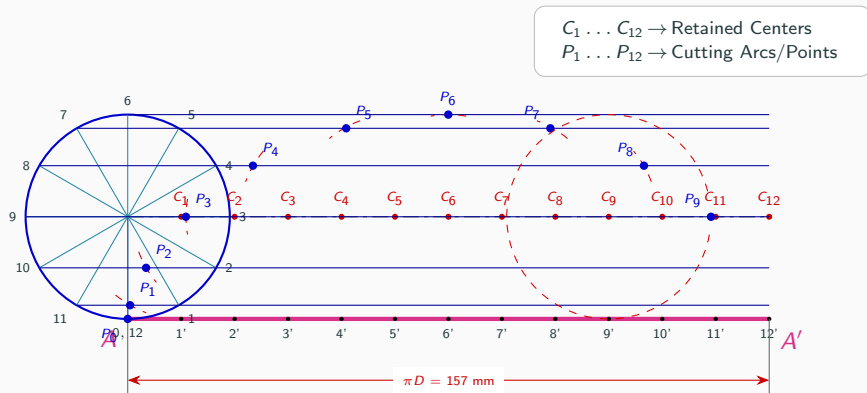
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



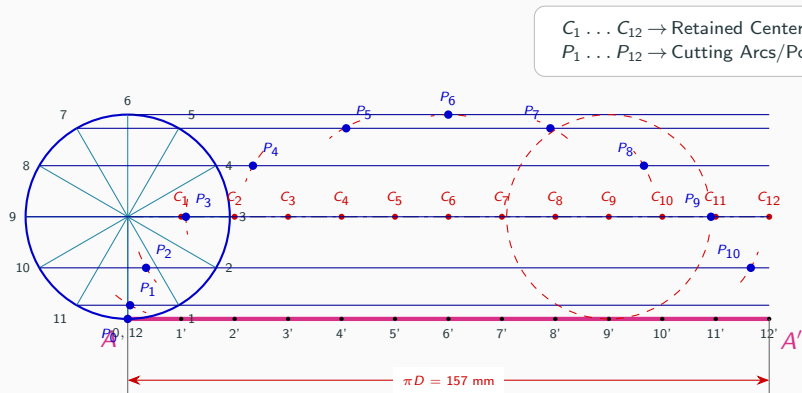
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



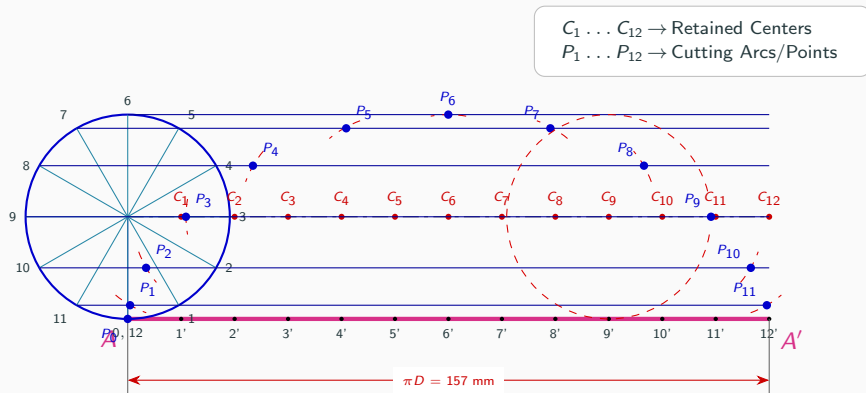
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



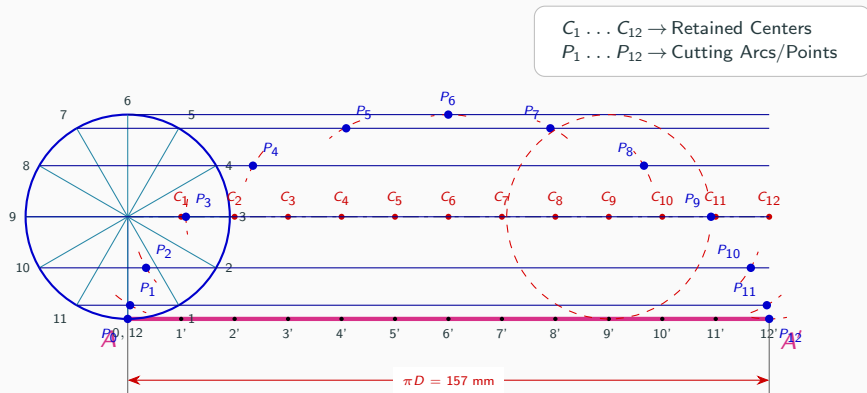
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



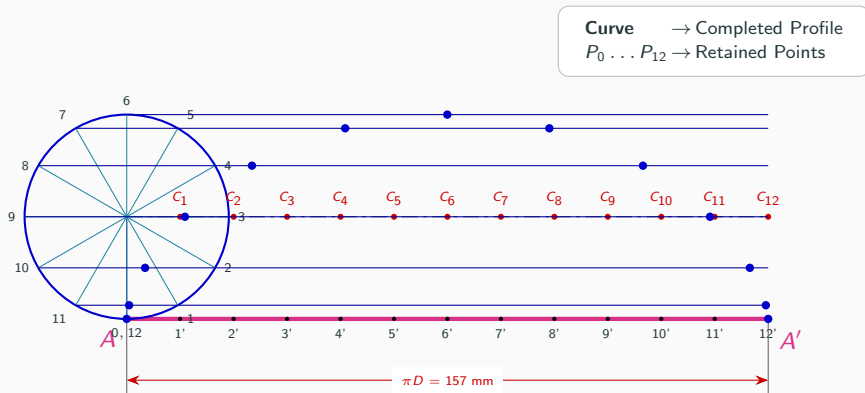
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



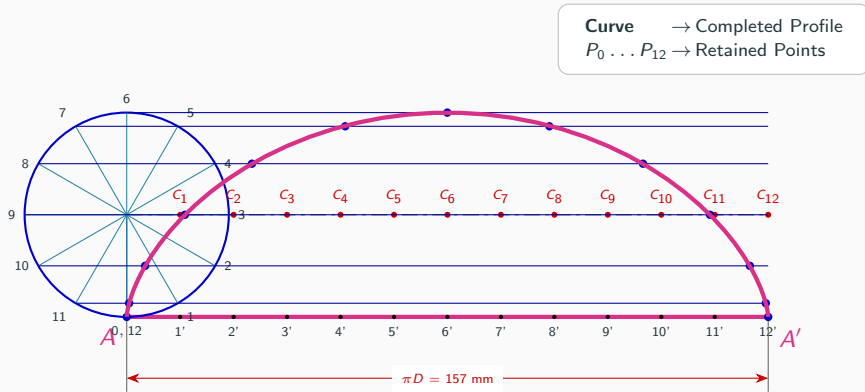
Step 6: Strike Cutting Arcs to Locate Points $P_1 \dots P_{12}$



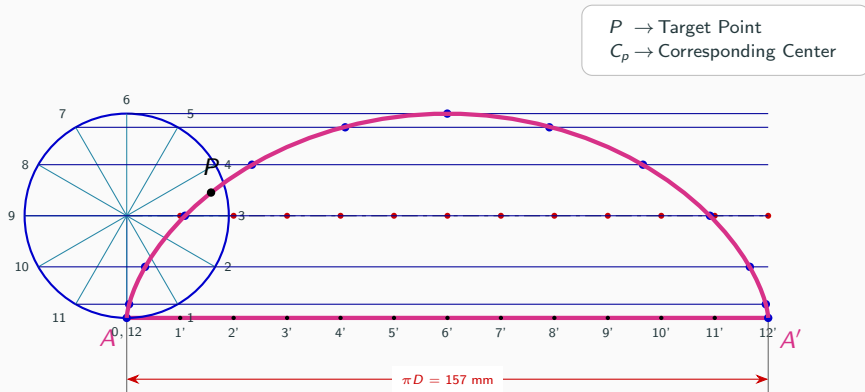
Step 7: Smoothly Join Points to Form Cycloid Curve



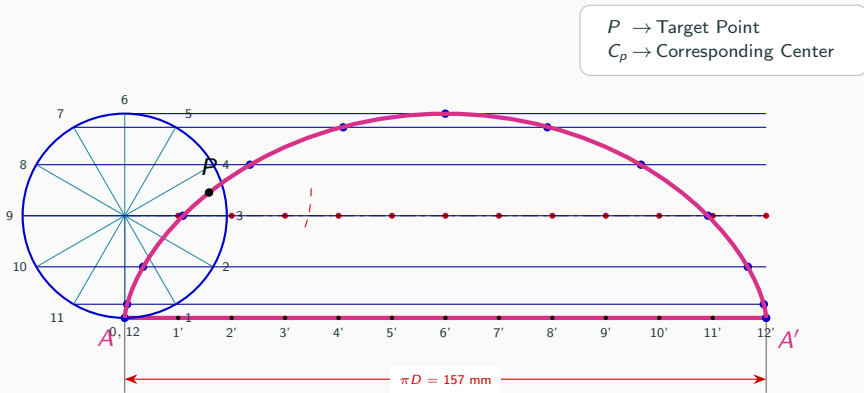
Step 7: Smoothly Join Points to Form Cycloid Curve



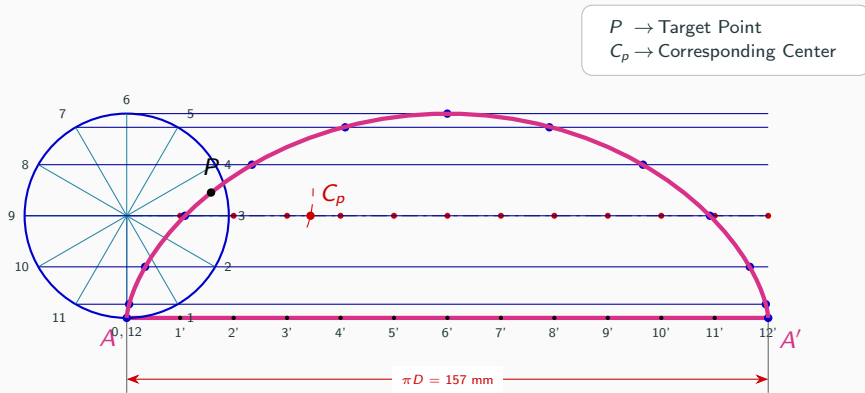
Step 8: Mark Point P & Locate Center C_p



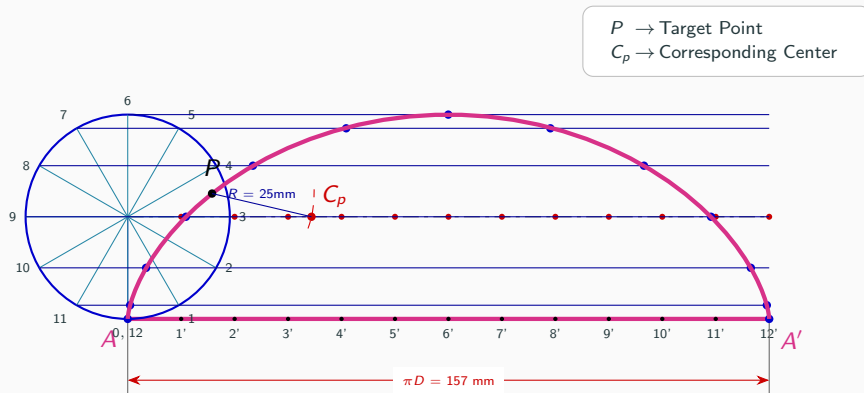
Step 8: Mark Point P & Locate Center C_p



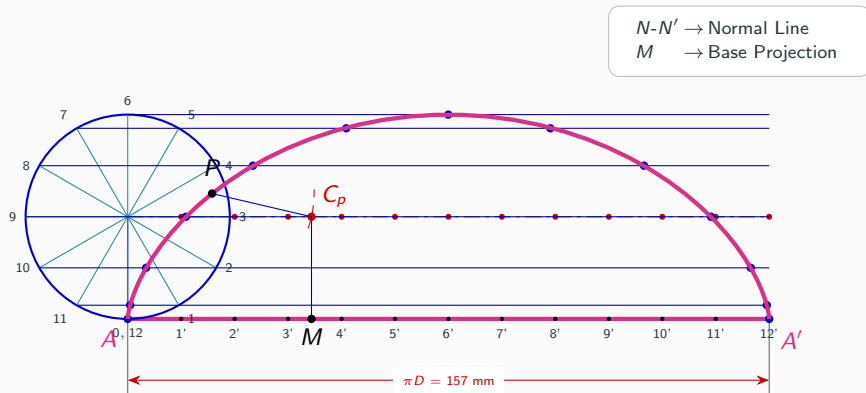
Step 8: Mark Point P & Locate Center C_p



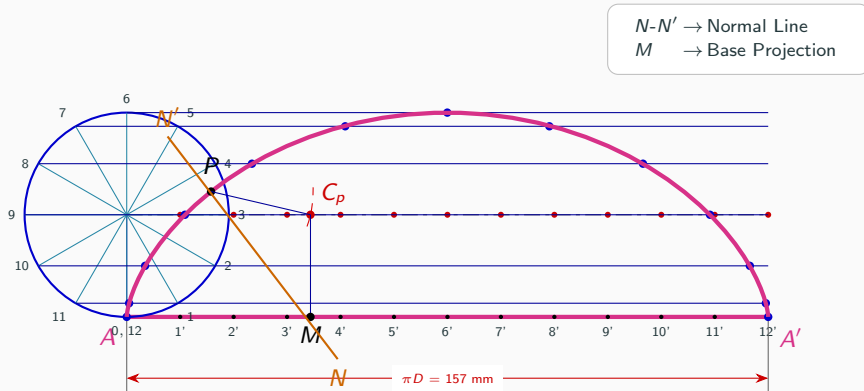
Step 8: Mark Point P & Locate Center C_p



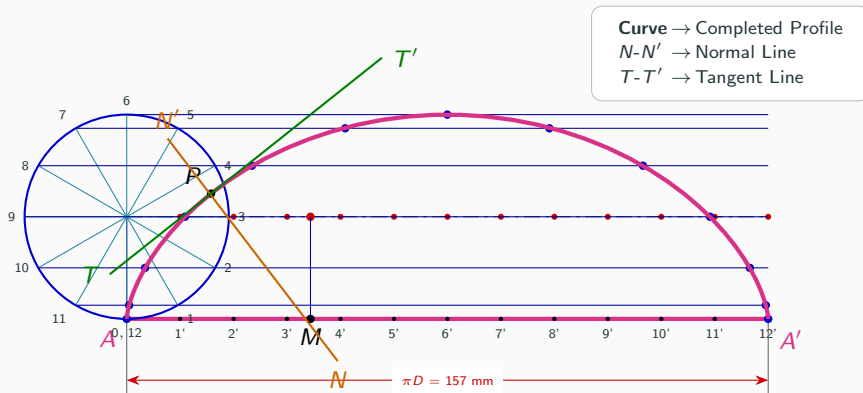
Step 9: Construct Normal Line $N-N'$



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Step 10: Construct Tangent Line $T-T'$ & Final Result



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